

A Novel Self-Calibrated UWB based indoor localization systems for context-aware applications

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Abstract—Location information is the most crucial information used in context-aware applications, e-commerce and IoT-based consumer applications. Traditional methods doesn't focus on network coverage, accuracy, hardware cost, and noise in dense environment. To defeat these issues, this paper presents a novel localization algorithm for UWB nodes adopting self-calibration and ToA measurement for context-aware applications. The Link quality induction values are used instead of RSSI for distance estimation by costing technique. A calibration factor (CF) is further introduce to automatically update the location information in mobility. As the signal strength can be distorted heavily due to shadowing and multi-path fading, the localization is estimated in noisy condition and extended Kalman filtering (EKF) is applied to refine the node coordinates. Simulation results shows that the positioning error is decreased with an overall accuracy of 0.23m and standard-deviation of 0.76m.

Index Terms—Ultra-wide band (UWB), link-quality induction (LQI), wireless sensor network (WSN), context-awareness.

I. INTRODUCTION

THE concept of Ambient Intelligence emerge the new ways and advancement in sensing and communication through context-awareness. The wireless personal network [1] and future home networks [2], are the key examples of context-aware European projects with mobility. The context-aware solutions utilize geographical locations, hardware and device information, and keep the historical record of usage along with environmental information. Ultra wide-band (UWB) based on IEEE 802.15.4z-2020 HRP ERDEV (enhanced ranging devices), gain much interest because of its

potential in many context-aware applications [3]. UWB performs better for short-range communications even in presence of multipath and dense environment due to its fine-delay resolution features. Over a huge bandwidth of ($> 500MHz$) an ultra short-pulses are transmitted in < 1 nanosecond of time period with frequency range of $3.1GHz$ to $10.6GHz$, which reduce the power consumption. UWB low power utilization and immunization to interference from other signals (different radio spectrum and signal type), secure ranging, high precision, and signal penetration, made this a striking technology for 3d and mobile-aware indoor localization [4].

A. Background

A growing interest on precise indoor localization has been greatly observed in recent years. In the era of e-commerce and consumer electronics, the demand for precise and efficient localization in sensor networks has become increasingly pronounced. The integration of location-aware technologies employing in different applications investigated enhancing user experience, enabling location-based services, and supporting a myriad of applications in the e-commerce and consumer electronics domains such as optical, acoustic, and radio frequency (RF) methods. The techniques based on RF may adopted continuous waves (CW) and impulse signals. The methods and technologies using CW are highly affected due to low accuracy, large network, and multipath scenario. Therefore, UWB technology became commercially available in late 1990's is a promising solution for indoor environment. The three main applications: 1) Sensors and communication; 2) tracking, navigation and localization; and 3) radar's are using the UWB because of real-time high precision [5].

UWB use time of flight (ToF) rather than received signal strength (RSSI) to compute distance. The well-known ToF based estimation methods includes angle of arrival (AoA), time of arrival (ToA), time difference of arrival (TDoA), RSSI, and received signal strength difference (RSSD) [6]. ToA and TDoA techniques requires accurate clock synchronization whereas AoA-based techniques necessitate precise antenna array. While the low cost and low complex techniques RSS and RSSD utilize signal strength for positioning estimation. ToA based techniques are more popular in many static and mobile based techniques which adopted the transmit time between transmitter and receiver for distance computation. The distance estimation in ToA can be affected by noise, multipath components (MPC's), propagation velocity between

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affected by Gaussian, or Naive noise intensity, because NLOS paths and errors can't be computed in time-varying scenario resulting in high localization error with low network coverage. The idea in [7] based on channel modeling and peak-detection is a best way to identify NLOS path in dense condition. In localization a trusted node transmit data packet with a ratio of transmission to received packets. The minimum acceptable packets are threshold. The node rating is computed by E-commerce formula commonly known as Sporas formula [8].

B. Motivation and Contribution

Motivated by the dense features of consumer electronics and e-commerce, this research enhance the sensor localization, envisioning a future where seamless integration enhances supply chain, logistics, and customer experience on IoT/ consumer electronics devices. We analyzed wide range of UWB positioning systems confirming that the tagged nodes are localizable in LOS conditions with decimeter (dm) level accuracy but the accuracy falls in case of NLOS environment. The poor findings in distance measurements in NLOS environment motivate us to further work on this area to proposed a better solution which even works on NLOS conditions. Furthermore, the level of precision can enhance context-awareness through real-time and complete location information.

After conducting a detailed survey on context-aware and UWB based localization it has been observed that in most techniques a proper node deployment has not been considered. Also for mobile based solution where context-awareness is crucial RSS is not a best choice because of weak signal and node failure. Therefore, we are using the LQI values for localization measurement. In the context of UWB technology, the adoption of LQI over RSSI holds significant importance. UWB systems, classified by their ability of data transmission over a broad spectrum with low power consumption, benefit from LQI's focus on assessing link quality comprehensively. Unlike RSSI, LQI considers factors such as interference, signal-to-noise ratio, and provides an accurate insight of the communication channel's reliability. The stability of LQI values, its standardization, and its resilience to environmental fluctuations in wireless sensor networks, including those utilizing UWB protocols, make it a preferred metric. To the best of our knowledge it is the first analytical and state-of-the art work for UWB based localization which combine the measurement of CF and then compute the UWB node localization using LQI values. We also differentiate between collaborative positioning and self-calibration which are only discussed for other technologies like wi-fi, Zigbee and Bluetooth. A set of nodes, beacons and gateways is deployed in this work based on UWB standard, with $\dot{n}$ nodes, $\dot{b}$ number of beacons, and $\dot{g}$ gateways, where $\dot{n} \gg [\dot{b} \ \& \ \dot{g}]$. The accuracy is depended on the beacons and gateways density, floor-plan and UWB tags. Beacons and gateways are fixed at the known positions while UWB sensors with unknown position and mobility. Nowadays sensor nodes also gain power through renewable or harvesting energy, e.g. from solar or vibrations, and radio waves, may having single radio. Beacons and gateways with line power supply, high performance CPUs, and capable to perform computation

(distance, LQI), within its range. The major contributions of this work is as follows.

- 1) We proposed the idea of context-aware localization in LOS and NLOS environment, where interference doesn't taken into account.
- 2) For positioning LQI values are used in order to measure the node position precisely.
- 3) The CF along with weight expression is introduced, so the nodes can calibrate their changes automatically.
- 4) LE is computed by using the values of distance affected by the noise, with the help of centroid formula.
- 5) The positioning accuracy is being compared with the well-known existing algorithm from literature.

The rest of the paper is organized as follows. A state-of-the-art UWB based techniques is presented in Section II. Problem statement and system model is presented in Section III, while proposed design is presented in Section IV followed by the numerical and analytical analysis in Section V. The Section VI conclude the proposed technique.

II. RELATED WORK

The triangle of sensor positioning, consumer electronic, and e-commerce is a dynamic research area with multiple implication. In order to engage UWB technology for indoor positioning, several localization techniques have been presented, for IoT/ consumer electronic devices where information of the known positions is extracted through radio signals drifting among reference points (anchor or beacons) and target nodes (unknown). UWB based positioning algorithms are classified into AoA, ToA, TDoA, RSS and RSSD; the localization techniques, further classified into range-free and range-based techniques as shown in Fig. 1.

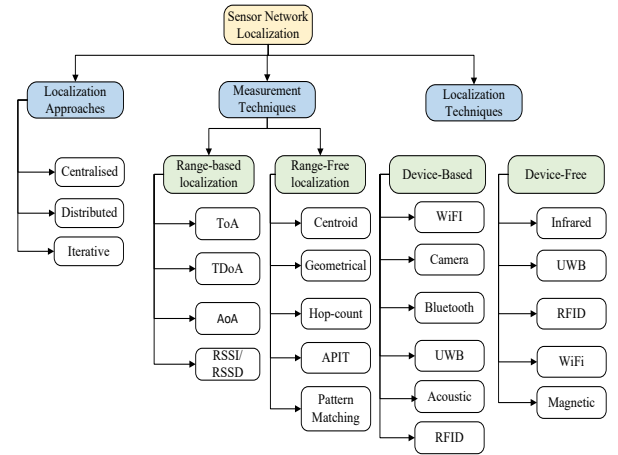


Fig. 1: Classification of localization algorithms.

In AoA techniques, signal reception angles is computed from signal amplitude or carrier phase over various antennas, then the location is estimated from junction point of the angle line. Localization algorithms based on AoA are highly sensitive to noise, fading, and NLOS resulting in low accuracy and high complexity. The geometry of antenna has a main role in positioning computation of AoA based algorithms, so

the increase in distance between transmitter and receiver may further decrease accuracy. In [9], localization is performed for centimeter distance estimation on smart UWB nodes. The work presents the in-depth study of AoA UWB measurement using low-power, compact solution integrating a novel module with phase difference of arrival (PDoA). Simulation shows the achievement of cm level accuracy along with 2.4° average angular accuracy.

A ToA based method, considering empirical approach based on machine learning is presented in [10], which classify channel condition (LOS and NLOS) in which information coming from LOS paths and estimates the mobile terminal's location. The work is compared with other state-of-the art methods and shows 83% improvement in localization accuracy. The author's in [11] design a positioning algorithm for coal mine which locate the workers in case of emergency using UWB technology. A monolithic chip set for UWB for indoor localization has been presented in [12] for LOS scenario with an accuracy of 8.3cm . The positioning accuracy was examined in [13] for UWB tags and improved localization accuracy in multipath environment with short transmitted pulses. [14], reduce the NLOS error with maximum likelihood estimation (MLE) and biased kalman filtering (BKF). On the other-side, a large scale LPS-LOS system based on ToA were presented in [15] with high precision and network coverage.

TDoA is also a promising method which is a time difference technique between sender and three or more receiver to form a triangulation. In [16], author proposed FlexTDoA using UWB radios in which user devices are remain passive and measure localization by listening to the communication among fixed anchors nodes using flexible time-division multiple-access (TDMA) scheme where time is divided in slots. The effect of physical speed is simulated against FlexTDoA and classic TDoA approach. The results reflects that FlexTDoA gain high accuracy of upto 38% reduction in error as compared to classic approach. The author's in [17], presents one-dimensional system having one tag and two sensors and use traditional TDoA method for position computation. For collision avoidance on-off keying (OOK) was used for tag synchronization, which improve the localization accuracy and power consumption. Similarly, another method proposed in [18], developed a multi-tag UWB network modulation that used minimum-shift keying (MSK) having range of 100m , $(1 - 20)\text{Hz}$ refresh-rate, and frequency range of $(2.4 - 2.48)\text{GHz}$ -($5.4 - 10.6$) GHz . A detailed study were presented in [19], about UWB localization with different complexities and accuracy requirements. A 'cm' accuracy were recorded in 3D environment that uses TDoA technique and four receivers for triangulation's. Path overlapping and angle dependent signal distortion is a main challenge for this technique.

A UAV landing mobile platform along with GPS is presented in [20]. Four UWB anchor nodes are deployed on the region of $(0.8 \times 1.4)\text{m}$ rectangular UVs landing platform. The coordinates are measured using linear least square (LLSs). The method provides an overall improvement of 61% as compared to other methods adopting least squares for error reduction.

A novel UWB indoor tracking system was presented in [21], having two sandwich systems; a mobile robot (MR) and a

control system. The first signal was detected by a dynamic threshold crossing technique which utilize hybrid mixture of ToA and TDoA for location computation.

III. SYSTEM MODEL

The integration of e-commerce, consumer electronics, and sensor localization introduces unique opportunities and challenges. The proposed system consider those challenges so that the system must be useful for customized applications. Our model provide a novel solution, meeting the requirements of IoT based consumer applications and e-commerce industry. A list of acronyms used in this paper is presented in Table I. Assuming that the receiver and transmitter are synchronized;

TABLE I: List of Acronyms.

Abbreviation	Definition
Υ_i	Channel attenuation
$\Psi(t)$	IR-WSB signal
ρ	Number of multipath components
$\zeta(t)$	Gaussian noise
η	Path loss exponent
ϱ	Distance from anchor to unknown node
$\wp$	Error reduction factor for ToA computation
$\dot{b}_i \mapsto \varrho_i$	Distance from a beacon to a node
$\dot{b}_k \mapsto \dot{g}_j$	Distance from a beacon to gateway
$\dot{n} \mapsto \varrho_i$	Distance from a node to node
$v(t)$	Waveform of a broadcasted pulse $p(t)$
$*$	band-pass code defined by the IEEE 802.15.4a protocol
$\mathfrak{U}_m$	Factor including the energy and path in vector form
$\hat{\omega}$	UWB tag estimated location
$[U]$	Decomposition matrix
$\Re_{i,j}$	Calibration factor
$\varkappa$	Node density
ϖ	Unidirectional frequency [constant]

thus, the impulse radio IR-WSB signal at the receiving end can be modeled under NLOS, multipath environment, and channel attenuation. The ranging model is defined by:

$$\Psi(t) = \sum_{i=1}^{\rho} \Upsilon_i p(t - \Upsilon_i), \quad (1)$$

where ρ represents the number of multipath components, $p(t)$ is the first signal, Υ_i represents channel attenuation. For ToA measurement, $\Upsilon_1 = \Upsilon_{ToA}$ for further analysis. Adding Gaussian noise, Eq. (1) can be further expressed as:

$$\Psi(t) = \sum_{i=1}^{\rho} \Upsilon_i p(t - \Upsilon_i) + \zeta(t), \quad (2)$$

where $\zeta(t)$ is a noise. Beacons and UWB tags are close to gateways, with longer LOS distance that affect ToA estimation.

$$\sqrt{(x - \hat{x}_i)^2 + (y - \hat{y}_i)^2 + (z - \hat{z}_i)^2} = \varrho_i - \eta_{i,j} \quad (3)$$

where:

$$\eta_{i,j} = \begin{cases} \zeta_{i,j} & \text{LOS;} \\ \zeta_{i,j} + \wp_{NLOS}^{\rho} & \text{NLOS;} \end{cases} \quad (4)$$

Here, $\wp$ is for the simulation for error and ϱ is a distance. Due to complexities in AWGN the probability density function (PDF) with Rayleigh fading $f(x) = \frac{x}{\sigma^2} \exp(-\frac{x^2}{2\sigma^2})$ is:

$$F_{\eta_{i,j}}^{LOS} = \frac{1}{\sqrt{2\pi\sigma_{i,j}^2}} \exp\left(-\frac{\eta_{i,j}^2}{2\sigma_{i,j}^2}\right) \quad (5)$$

$$F_{\eta_{i,j}}^{NLOS} = \frac{1}{\sqrt{2\pi\sigma_{i,j}^2}} \exp\left(-\frac{(\eta_{i,j} - \mu_b)^2}{2\sigma_{i,j}^2}\right) \Rightarrow \Psi(t) \quad (6)$$

$$F_{\eta_{i,j}}^{NLOS} = \frac{1}{\beta} \left[\Upsilon_i \left(\frac{\eta_{i,j} - n_{max}}{\sigma_{i,j}} \right) - \Upsilon_i \left(\frac{\eta_{i,j} - n_{min}}{\sigma_{i,j}} \right) \right] \quad (7)$$

where ϱ_i is a distance, NLOS fractional complex noise β computed by the *max* and *min* of normalized distance with a commulative distribution function (CDF) of channel attenuation, where noise factor $\zeta(t)$ satisfy $\zeta_{i,j} \sim \eta(0, \sigma_{i,j}^2)$.

$$\rho_{\eta_{i,j}}^{NLOS} = \frac{\chi}{2} \exp\left[-\chi \left(\eta_{i,j} - \frac{\chi^2 \sigma_{i,j}^2}{2} \right) \tilde{\delta} \left(\frac{\chi^2 \sigma_{i,j}^2}{\sqrt{2\sigma_{i,j}^2}} \right) \right] \quad (8)$$

where χ is a positive constant, and $\tilde{\delta}$ is the positioning error distribution. The distance in coordinate form will be:

$$\dot{b} \mapsto \varrho_i = \|\omega - \omega_i\| = \sqrt{(x_\omega - \hat{x}_i)^2 + (y_\omega - \hat{y}_i)^2 + (z_\omega - \hat{z}_i)^2} \quad (9)$$

where $\dot{b} \mapsto \varrho_i$ is a distance from beacon to a node, and $(\hat{x}_i, \hat{y}_i, \hat{z}_i)$ and $(x_\omega, y_\omega, z_\omega)$ is a beacon and UWB tag position, respectively. Simplifying for the least square modeling,

$$\dot{b} \mapsto \varrho_i^2 = x_\omega^2 + \hat{x}_i^2 - 2x_\omega \hat{x}_i + y_\omega^2 + \hat{y}_i^2 - 2y_\omega \hat{y}_i + z_\omega^2 + \hat{z}_i^2 - 2z_\omega \hat{z}_i \quad (10)$$

Similarly the distance between nodes and anchor node will be:

$$\dot{n} \mapsto \varrho_i^2 = x_N^2 + \hat{x}_i^2 - 2x_N \hat{x}_i + y_N^2 + \hat{y}_i^2 - 2y_N \hat{y}_i + z_N^2 + \hat{z}_i^2 - 2z_N \hat{z}_i \quad (11)$$

$$= 2(x_N - \hat{x}_i)x_\omega + 2(y_N - \hat{y}_i)y_\omega + 2(z_N - \hat{z}_i)z_\omega \quad (12)$$

which can be written using a matrix relation in the form $k = T\omega$, and modeled in the algorithm design section.

IV. TOA ESTIMATION WITH TWO-STEP RANGING

A two-step range was adopted for the ToA estimation. Initially, the first signal path is detected using a bandpass filter (BPF), with bandwidth B and frequency f_o . The RSS, followed by the energy detector (Eq. (2)) can be written as:

$$s(t) = \sum_{m=0}^* v(t - \dot{n} \cdot f_o - mT) + \zeta(t), \quad (13)$$

where $v(t)$ is the waveform of a broadcasted pulse $p(t)$ and $*$ is a band-pass code. Additionally, $s(t)$ denotes the received signal with the channel impulse response. By adopting maximum likelihood estimation (MLE) there are $\dot{n} - 1$ parameters to measure, that is, the energy of each signal (described in Eq. (42) in Section VI-A) and the value of each path on a successful signal transmitter. These factors described in vector form $\theta^{(l)} = [\vartheta_{\dot{n}}, \vartheta_{\dot{n}+1}, \dots, \vartheta_{\dot{n}+m}, s(t)]$, with noise. The PDF is used as an argument in the MLE equation as follows:

$$s(t, \theta^{(l)}) = \prod_{m=0}^* F(\argmax(\vartheta_i) \cdot \vartheta_m, \Psi(t), \varrho \mapsto \zeta(t)) \quad (14)$$

$$\ln s(t, \theta^{(l)}) = \ln \frac{F(\argmax(\vartheta_i) \cdot f_o)}{s(t)} \ln \frac{l_{v/2}(\frac{\vartheta_i \vartheta_m}{cd})}{\vartheta_m} + \zeta(t) \quad (15)$$

$$\ln s(t, \theta^{(l)}) = \ln \frac{l_{v/2}(\frac{\vartheta_i \vartheta_m}{cd})}{s(t) \cdot \vartheta_m} + \zeta(t) \quad (16)$$

$$\ln s(t, \theta^{(l)}) = \ln \frac{l_{v/2}(\frac{\vartheta_i \vartheta_m}{cd})}{s(t) \cdot \vartheta_m} - \frac{\sum_{m=0}^n \vartheta_i}{f_o s(t)} \quad (17)$$

where ϑ_i denotes channel attenuation with time interval F , ϑ_m is a factor having energy and path in vector form. The derived function has small fraction, either positive or negative. In presence of multiple MPCs, the MLE may yield negative values, which is incorrect computation of ToA, and can be fixed using the MLE stabilization explained in Algorithm 1.

Algorithm 1 Time of Arrival Estimation.

- 1: Initializing a signal $\Psi(t)$ with noise $\zeta(t)$
 - 2: $\begin{cases} \zeta_{i,j} & LOS; \\ \zeta_{i,j} + \rho_{NLOS} & NLOS; \end{cases}$: decomposition to distance
 - 3: Peak signal: Determine the estimated distance from the UWB tag to the gateway.
 - 4: $\dot{b} \mapsto \varrho_i^2, \dot{n} \mapsto \varrho_i^2 + \Psi(t)$
 - 5: $F_{\eta_{i,j}}^{LOS}$ for direct path
 - 6: $F_{\eta_{i,j}}^{NLOS}$ for MPC $\Rightarrow \dot{b} \mapsto \varrho_i^2$
 - 7: $s(t, \theta^{(l)}) \Rightarrow [\vartheta_{\dot{n}}, s(t)]$ MLE components in vector form
 - 8: ML estimate from Eq. 43
 - 9: **return** $\Psi(t)$ [Maximum likelihood]
 - 10: PREDICT($\Psi(t), ML$)
 - 11: **return** $\varrho_{i,j} \Rightarrow Estimated \Psi$
-

V. UWB-LOCALIZATION USING CALIBRATION FACTOR

In this section, we presents a novel localization algorithm, which use LQI and use CF under mobility. The presented analytical structure varying levels of shadowing, including both correlated and independent shadowing environments, and also take inaccuracy factor while introducing calibration factor. In weighted centroid (WCL), beacons $\dot{b}(x, y, z)$ with a known position broadcast their ID and position coordinates to all UWB nodes $\dot{n}(x, y, z)$ in the neighbor using mere communication. WCL have many challenges including high maintenance cost, time consumption and system failure if there is no node in a particular region. In this case CF fails to execute further and therefore, its not energy-efficient. In a real scenario, weightings are introduced to scale the importance of beacons according to their distances from the nodes.

$$\dot{n}_i = \frac{\sum_{j=1}^n h_{i,j} \dot{g}}{\sum_{j=1}^n h_{i,j}} \cdot \Psi(t) \quad ; \quad h = \frac{1}{(\varrho_{i,j})^l} \quad (18)$$

where h is a weight. IEEE 802.15.4 specifies LQI values for packet error calculations via $Mgmt_{Lqi}$, in matrix form.

Because of similar nature of RSSI and LQI, the distance between node and gateways will be computed the same way.

$$\varrho_{i,j} = S_0 10^{-S_1 \cdot Q_{i,j}} \quad ; \quad Q_{i,j} = \frac{LQI_{i,j}}{LQI_{max}} \quad (19)$$

Eq. (19) is unsuitable under extreme conditions when $LQI=0$; thus, using the third-order polynomial $LQI_{i,j} = S_3 \cdot \varrho_{i,j}^3 + S_2 \cdot \varrho_{i,j}^2 + \dots + S_0$, we have

$$\varrho_{i,j} = \sqrt[n]{\frac{a}{Q_{i,j}}} \quad (20)$$

where a is a constant initial CF and n is a path loss component computed by $\varrho = \varrho_0 \cdot 10^{\frac{P_0 - P_{RX} + E_\sigma}{10n}}$, which lies between 2 and 4 in our case. In a multipath environment, the signals obtained from Eq. (1) can be expressed as

$$e(t) = \Psi(t) + \zeta(t) \quad (21)$$

$$h(t) = \sqrt{E_p} \sum_{i=1}^{\rho} \mathfrak{I}ip(t - \Upsilon_i) + \zeta(t) \quad (22)$$

where $\Upsilon_1^\rho, \Upsilon_2^\rho, \Upsilon_3^\rho, \dots, \Upsilon_f^\rho, \mathfrak{I}_1^\rho, \mathfrak{I}_2^\rho, \mathfrak{I}_3^\rho, \dots, \mathfrak{I}_f^\rho$ is an array of parameters with a delay of Υ_f^ρ and a $\mathfrak{I}_f^\rho$ path. E_p is a standard deviational error factor. The channel attenuation, distance, and LQIs are stored in a matrix as, which satisfies $k = T\omega$.

$$k = \begin{bmatrix} \varrho_1^2 - x_1^2 - y_1^2 - z_1^2 - \varrho_N^2 - x_N^2 - y_N^2 - z_N^2 \\ \varrho_2^2 - x_2^2 - y_2^2 - z_2^2 - \varrho_N^2 - x_N^2 - y_N^2 - z_N^2 \\ \vdots \\ \varrho_{N-1}^2 - x_{N-1}^2 - y_{N-1}^2 - z_{N-1}^2 - \varrho_N^2 - x_N^2 - y_N^2 - z_N^2 \end{bmatrix} \quad (23)$$

$$T = 2 \times \begin{bmatrix} x_1 - x_N & y_1 - y_N & z_1 - z_N \\ x_2 - x_N & y_2 - y_N & z_2 - z_N \\ \vdots & \vdots & \vdots \\ x_{N-1} - x_N & y_{N-1} - y_N & z_{N-1} - z_N \end{bmatrix}; \omega = \begin{bmatrix} x \\ y \\ z \end{bmatrix} \quad (24)$$

Now, the estimated position of the beacons $\dot{b}'_k$ are compared with known locations along with ToA estimates.

$$\dot{b}'_k \mapsto \dot{g}_j = \frac{\sum_{j=1}^n h_{j,k} \cdot \dot{g}_j}{\sum_{j=1}^n h_{j,k}} \cdot \sqrt{E_p} \sum_{i=1}^{\rho} \mathfrak{I}ip(t - \Upsilon_i) \quad (25)$$

$$\dot{b}'_k \mapsto \dot{n}_r = \frac{\sum_{j=1}^n h_{j,k} \cdot \dot{n}_r}{\sum_{j=1}^n h_{j,k}} \cdot \sqrt{E_p} \sum_{i=1}^{\rho} \mathfrak{I}ip(t - \Upsilon_i) \quad (26)$$

Next phase is to measure the approximate position of the beacons $\dot{b}'_k$. The difference in the estimated values further helps to introduce a threshold known as the CF. The beacon-estimated error can be computed using the CL as follows:

$$\dot{b}_k[Error] = \dot{b}'_k \sqrt{(\dot{b}'_{xk} - \dot{b}_{xk})^2 + (\dot{b}'_{yk} - \dot{b}_{yk})^2 + (\dot{b}'_{zk} - \dot{b}_{zk})^2} \quad (27)$$

The beacon-positioning error is proportional to the error in the node position, and we can now combine Eq. (20) and a part of Eq. (18) to obtain a simplified weight function, i.e.,

$$h = \frac{a}{Q_{i,j}} \quad (28)$$

$Q_{i,j}$ has already been derived from Eqs. (19). The path loss and distance are now eliminated, leaving a constant 'a' derived from the LQI [22]. The estimated position of the tag can be calculated using the relationship $k = T\omega$: Let T be a square matrix obtained from Eq. (24), the UWB tag location is

$$\hat{\omega} = T^{-1}k \quad (29)$$

where $\hat{\omega}$ denotes a UWB tag estimated location of the UWB tag. The least-squares estimation (LSE) is applied to Eq. (29):

$$\hat{\omega} = T_{LSE}^{-1}k, \quad (30)$$

By substituting the LSE and inverse matrices, the power of ρ becomes $\hat{\omega} \approx [U]T^{-1}[U]k$, where $[U]$ is the decomposition matrix. The number of LOS components is smaller than the number of NLOS components, so the contents of LOS are:

$$\hat{\omega} = \arg \max_{1 \leq \rho \leq P} T_\rho^2. \quad (31)$$

Lemma 5.1: If fewer LQIs are missing, the estimated position cannot be computed using square of the matrix T : which is obvious if fewer LQIs are missing, and the desired numbers of rows and columns are insufficient to take the square of the matrix. In this case, a pseudo-inverse methodology is used to estimate the nodes.

$$\hat{\omega} = (T^T T)^T k \quad (32)$$

Considering that $\dot{b}$ and $\dot{n}$ are similar in terms of transmission power and losses, this constant remains the same for all devices within a sensing region. The constant factor a is eliminated by inserting the weight $h = \frac{a}{Q_{i,j}}$ into Eq. (25).

$$\dot{b}'_k \mapsto \dot{g}_j = \frac{\sum_{j=1}^n \frac{\dot{g}_j}{Q_{k,j}}}{\sum_{j=1}^n \frac{1}{Q_{k,j}}} \cdot \Psi(t) \quad (33)$$

where $\dot{b}'_k \mapsto \dot{g}_j$ is a distance from the beacon to gateway. Variations in the node position due to antenna direction, node failure, or radio changes lead to erroneous weighting measurements and low accuracy. Therefore, to update these changes, we define a CF $\mathfrak{R}_{i,j}$ that minimizes $\dot{b}'_k - \dot{b}_k$:

$$\dot{b}_k[Error] = \|\dot{b}'_k - \dot{b}_k\| \Rightarrow \left\| \frac{\sum_{j=1}^n \frac{\mathfrak{R}_{i,j} \dot{g}_j}{Q_{k,j}}}{\sum_{j=1}^n \frac{\mathfrak{R}_{i,j}}{Q_{k,j}}} \right\| \cdot \Psi(t), \quad (34)$$

which is used to calculate the CF. We need to analyze how the CF can be implemented at the nodes. For this, a UWB node can be part of at least two beacons and a gateway to form a triangulation. For a node, we can rewrite Eq. (34) with CF:

$$\dot{N}'_j = \left\| \frac{\sum_{j=1}^n \frac{\mathfrak{R}_{i,j} \dot{n}_j}{Q_{i,j}}}{\sum_{j=1}^n \frac{\mathfrak{R}_{i,j}}{Q_{i,j}}} \right\| \cdot \Psi(t) \quad (35)$$

AWGN is added to the distance equation. In the first ToA path, $\zeta_{i,j} \sim \eta(0, \sigma_{i,j}^2)$,

$$V_n = \varrho_i \zeta(t) (1 - \varrho_i \zeta(t))^{\eta(ToA + n, \sigma_{i,j}^2)} \quad (36)$$

$$V_n = \varrho_i (1 - \zeta(t))^{\eta ToA} \prod_{i=1}^{n-1} (1 - \varrho_i) \quad (37)$$

$$PDF | \eta ToA = 1 - \prod_{i=1}^{n-1} (1 - \varrho_i) \Rightarrow \begin{cases} \dot{b}'_k \mapsto \dot{g}_j \\ \dot{b}'_k \mapsto \dot{n}_r \end{cases} \quad (38)$$

$$F\{(\hat{\Upsilon}_i - \Upsilon_i) | \eta ToA\} = \sum_n F(\hat{\Upsilon}_i - \Upsilon_i) |_{max} + F(\hat{\Upsilon}_i - \Upsilon_i) |_{min} \cdot \Psi(t) \quad (39)$$

where V_n is an estimated distance between the nodes and gateways. The maximum and minimum LQI values are used in the centroid formula to measure the localization error (LE).

$$\frac{\mathfrak{R}_{i,j}}{\varrho_i[avg]} = \Psi(t) \times \sum_{k=1}^n \frac{\mathfrak{R}_{j,k}}{\varrho_{i,k}} \quad (40)$$

$$l_{\varrho_{i,k}} = \sqrt{(\dot{N}_{i,x} - \dot{b}'_{k,x})^2 + (\dot{N}_{i,y} - \dot{b}'_{k,y})^2 + (\dot{N}_{i,z} - \dot{b}'_{k,z})^2} \quad (41)$$

Here, $l_{\varrho_{i,k}}$ denotes the average LE. Algorithm 2 describes the procedure for node localization and the derivation of the CF.

The proposed solution is novel in several aspects: First the adoption of LQI values. Second, the proposed technique assumes that the CF used to minimize the beacon-positioning error are identical for all nodes. Therefore, if the CF changes for the beacons, it changes for the nodes in the same manner. Third, there is no direct communication between the beacons and nodes. Everything passes through the gateways. This implies that the gateways perform almost all the work, which save considerable energy for the nodes. This is another area in which the proposed algorithm is superior. In practical scenarios, depending on the system administrator and requirements, recalibration can be performed every 15 min, every 60 min, or whenever $\|\dot{b}'_k - \dot{b}_k\|$ exceeds a certain threshold.

VI. RESULTS AND DISCUSSION

In this section, we describe the performance of the proposed technique. We attempt to simulate as realistic a case as possible, that is, one representative of a typical office or factory environment. We assume an area of $150m \times 70m$ with 7 beacons, 10 UWB nodes, and 7 gateways, with a reference node in the right corner along with the NLOS placement, as shown in Fig. 2. The realization was achieved with a sample rate of 10 GHz and a bandwidth of 4-12 GHz, with a uniform distance of $[1 - 4]m$.

A. RMSE to SNR

The root mean square error (RMSE) was compared with those of a few other well-known techniques, including the maximum probability of detection and constant false alarm rate for the radar theory presented in [23]. The fixed threshold was adopted [24], with the strongest path having the highest repetitive LQI values. The design parameter, which is used as

a normalized threshold, is set to 0.2, and for a false alarm, the rate is between 2.0 – 2.5 as per the CF requirement. After applying bandpass filtering to Eq. (2), the signal received comprises different pulses (extracted using an energy detector).

$$\mathfrak{U}_{\dot{n}, \dot{b}} = \frac{(e_n^\rho - 1)e^{-\rho}}{e_n/\rho(e^\rho - 1)} \prod_a^b \Psi(t) \quad (42)$$

$$\therefore ML \text{ estimate } \hat{\sigma}_{\mathfrak{U}}^2 = \frac{1}{2\mathfrak{U}_{\dot{n}, \dot{b}}} (\Psi(\hat{t}) - (\varrho_i - \eta_{i,j})) \quad (43)$$

Algorithm 2 Localization and UWB data fusion using CF.

- 1: Deployment $\leftarrow$ office/ factory environment with LOS and NLOC components.
 - 2: n number of nodes $\dot{n}$.
 - 3: b number of beacons $\dot{b}$.
 - 4: g number of gateways $\dot{g}$.
 - 5: noise $\zeta(t)$, which is a part of the distance.
 - 6: Receive signal strength in the form of LQI. $\leftarrow Mgmt_{Lqi}$.
 - 7: $\mathbf{RSS} \leftarrow \frac{LQI_{i,j}}{LQI_{max}}$
 - 8: if $LQI = 0$.
 - 9: $LQI_{i,j} = S_3 \cdot \varrho_{i,j}^3 + S_2 \cdot \varrho_{i,j}^2 + \dots + S_0$ including distance
 - 10: $k = T\omega$ matrix of all LQI values.
 - 11: Else return $\mathbf{RSS}$.
 - 12: Broadcast signal from $\dot{b}$ to $\dot{g}$ and $\dot{n}$.
 - 13: First path signal and ToA from Algorithm 1.
 - 14: Estimated beacons position $\dot{b}_k[Error]$.
 - 15: $\dot{b}_k[Error] = \sqrt{(\dot{b}'_{(x,y,z)k} - \dot{b}_{(x,y,z)k})^2}$.
 - 16: weight function $a/Q_{i,j}$ pathloss eliminated.
 - 17: Consider LQI matrix: if T a square matrix.
 - 18: $\hat{\omega} = T^{-1}k$ for LOS states.
 - 19: $\hat{\omega} = (T^T T)^{-1} T^T k$ in case T not a square matrix.
 - 20: $\dot{b}'_k \mapsto \dot{g}_j = [\sum_{j=1}^n \frac{\dot{g}_j}{Q_{k,j}}] / [\sum_{j=1}^n \frac{1}{Q_{k,j}}] \cdot \Psi(t)$ weight and PL is a part of transmission: Eq: 33.
 - 21: **return** $h[weight], distance$.
 - 22: [find a maximum and minimum point from LQI matrix].
 - 23: [Computation of region for network coverage].
 - 24: [Nodes with higher/ repetitive RSS are selected as LQI].
 - 25: derivation of CF. $\| [\sum_{j=1}^n \frac{\mathfrak{R}_{i,j} \dot{g}_j}{Q_{k,j}}] / [\sum_{j=1}^n \frac{\mathfrak{R}_{i,j}}{Q_{k,j}}] \|$
 - 26: distance $\leftarrow \zeta_{i,j} \sim \eta(0, \sigma_{i,j}^2) + \varrho_i$
 - 27: [Node positioning is the same as beacon positioning].
 - 28: [CF included in distance equation], Eq 40.
 - 29: [Localization error: $\sqrt{((\dot{N}_{i,[x,y,z]} - \dot{b}'_{k,[x,y,z]})^2}$.
 - 30: [Estimated node $\dot{n}$ position].
-

Following the acquisition of ToA, a part of $\Psi(t)$ with a time interval $[p(t) + i\Upsilon_i p(t), (i+1)v_i p(t+1)]$ is derived, along with the first path detection and the next interval when the beacon again transmits its packets, illustrated in Fig. 3. The energy detector output was extracted using $\frac{p(t=1)}{p(t)[first \ path]}$. The RMSE for different channels in LOS and NLOS environments were compared for different SNRs, as shown in Fig. 4, and was determined based on the means of 5000 trials. The proposed ToA method obtained better results when we receive

a relative number of LQI values. The constant-rate alarm and maximum probability methods achieved worse performances when $SNR < 0dB$. Thus, having a fixed calibration factor, which is a constant a in our algorithm, will help effectively measure the ToA under LOS and NLOS conditions.

B. Implementation of Calibration Factor

The implementation of the CF and LE computation begins after ToA measurement with centroid determination. In CF implementation, we have $\dot{n}$ nodes with unknown positions, a number of gateways with known positions $\dot{g}_j$, and a number of beacons with known positions $\dot{b}$. The gateways should be placed so that the angular differences between the nodes and beacons in the measurements are not very narrow. Steps for real-world implementation of the proposed solution are:

- 1) The system computes the LQI from each beacon to the gateway in sensing region. The relative $Q_{k,j}$ can be obtained from $LQI_{i,j} = S_3 \cdot \varrho_{i,j}^3 + S_2 \cdot \varrho_{i,j}^2 + \dots + S_0$.
- 2) The initial beacon position error $\dot{b}_k[Error]$ can be computed using the self-calibration parameter, $\mathcal{R}_{i,j} = 1$.
- 3) We now minimize $\dot{b}_k[Error]$ by tuning the CF.
- 4) Amplified UWB systems can have a range of 200 m, whereas non-amplified, attenuated systems might have a range of 1 – 50 m. Following are the steps to minimize calibration equation to decrease complexity.
 - a) Instantiate an array of $\dot{g}$, $\dot{b}$, and $\dot{b}'_k$. In addition, initialize the matrix of $Q_{k,j}$ and $\mathcal{R}_{i,j}$, which goes from $Q_{11} \rightarrow Q_{ij}$ and $\mathcal{R}_{11} \rightarrow \mathcal{R}_{ij}$.
 - b) Assuming that, B_1 is at LOS to all gateways, the LQI values close to each gateway are obtained.

$$\dot{b}'_1 = \left\| \frac{\sum_{j=1}^g \frac{1 \times \dot{g}_j}{Q_{1j}}}{\sum_{j=1}^g \frac{1}{Q_{1j}}} \right\| \cdot \Psi(t) \quad (44)$$

- c) Subsequently, all data are known, and thus the error is finally computed through Eq. (34).
- d) Compute $\mathcal{R}_{i,j}$ for an error reduction by Eq. (40). This applied in a loop in which the calibration equation is recalculated and every new $\dot{b}_k[Error]$ is compared with the previous one. For each CF, the adjustment that gives a lower $\dot{b}_k$ is kept.
- e) AWGN $\zeta(t)$ is added to the distance. Despite knowing the beacon's position, we have added this step to check to authenticate the proposed CF.
- f) After estimating node positions, we can now estimate the distance using Eq. (41). If there is only one or no UWB nodes, the CF will set in same manner as for the beacon and nodes to gateway.

$$\therefore \mathcal{R}_{ij} \Leftrightarrow \begin{cases} \dot{b}'_k \mapsto \dot{g}_j \\ \dot{b}'_k \mapsto \dot{n}_r \end{cases} \text{ same constant value} \quad (45)$$

The proposed scheme ensures minimal computational burden on nodes and beacons, enabling them to be small and battery-driven. This flexibility leads to an improved

and efficient implementation compared to other range-based systems."

C. Localization Error (LE) (Numerical Results)

The proposed algorithm was first tested numerically using a CL algorithm. We observed that in the CL, the nodes could compute their estimated locations based on the neighbor beacon nodes. The average LQI is recorded using distance function as shown in Table II.

$$\varrho_i = 210.10^{-3.96 \frac{LQI}{255}} \quad (46)$$

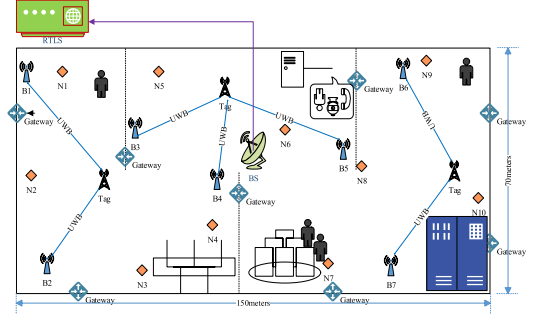


Fig. 2: Simulation area of $150m \times 70m$ is size.

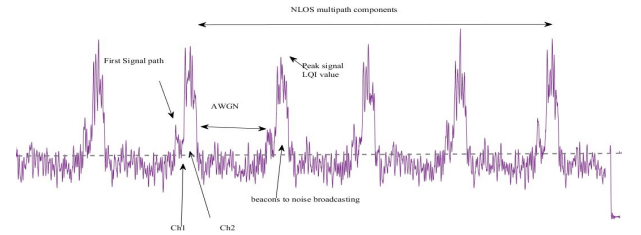


Fig. 3: Comparing RMSE-SNR with different ToAs in LOS and NLOS scenarios.

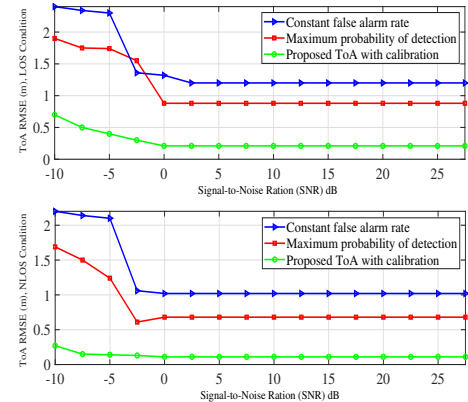


Fig. 4: Comparing RMSE-SNR with different ToAs under LOS and NLOS scenarios.

To calculate the error in changing environment, a distortion function is added similar to one proposed by [25].

$$\varrho_i = 210.10^{-3.96 \frac{LQI}{255}} \cdot (1 + a_o \cdot \sin(\pi \cdot t)) \quad (47)$$

Initially, LQI are not considered for CL implementation thus, a fixed RSS value was applied. From Fig. 4, N_1 can be seen by three beacons, B_1, B_2 , and B_3 , resulting in different LQI values. The LE was computed for average, LOS, and NLOS, as shown in Table III. An overall mean error of 2.64 m for the average condition and 2.59m, and 3.2m under LOS and NLOS environments, respectively, were recorded. To overcome the absence of nodes and network coverage, the weight was used as a distance mark, providing a precise localization. The same LQI values were used to verify the robustness of the weighted algorithms. With a fixed path loss exponent of 2, we obtained the smallest LE, as shown in Table IV. In [22], an improved accuracy metric was used that is normalized through the transmission range R . The MLE for the average condition was 1.28m and is 1.22m, and 1.54m for LOS and NLOS, respectively. The MLE was 1.31m for the WCL, as shown in Fig. 5.

For LE with CF, a threshold of $[-100 \text{ to } 100]$ was selected as the CF. The selected CF must satisfy the relation $\|\dot{b}_k - \dot{b}_k\|$. With the CF having favorable properties, such as in (1), if there is no beacon or node failure, the CF is $CF = 0$, (2) the network coverage, and (3) the accuracy of the proposed system directly affected the beacon and node positions. The calculated CF $\mathcal{R}_{i,j}$ for nodes $\dot{n}$ and beacons $\dot{b}$ computed from Eq. (40) are listed in Table V. Owing to symmetry, these new values are used in the node-calibrated relation in Eq. (38), to compute the estimated locations of the unknown nodes. To test this phenomenon, we deploy 5000 nodes in the 3D area. The estimated node locations and plotted diagrams are presented in Table VI and Fig. 5(a). The mean error using the CF in the presence of noise even decreases to 0.49 m, 0.33 m, and 0.59 m for the average, LOS, and NLOS conditions, respectively, which shows the high precision accuracy for UWB nodes localization where the transmission range is minimal. The total area of consideration was $150 \text{ m} \times 70 \text{ m}$, with 5000 trials; thus, the total distance covered in the simulation is:

$$\varrho_i = \sqrt[3]{(1158 \times 10^3) + (5 \times 10^3)} \quad (48)$$

Through increasing node density, we noted that LE decrease instantly, even in presence of noise. The beacon node density is defined by $\dot{b}/(\dot{n} + \dot{b})$. If we divide the whole area in a region, then the % density is computed by the following relation.

$$\varkappa(\%) = \frac{\Gamma}{R \cdot \dot{b}/(\dot{n} + \dot{b})} \quad (49)$$

where $\varkappa$ is a node density in %, Γ is a communication radius. If the total number of $\varkappa$ is less than the number of gateways in a region, the LE can't obtained LQI values in several areas. It can be observed from Fig. 5(b) that in case of 10% beacons the localization error lies between $(1.2 - 2.9)m$ with higher % standard deviation because the data elements are scattered around the region. By increasing beacon nodes density the error falls between $(0.2 - 0.89)m$.

$$P_{analysis} = \frac{\sum_{i=1}^N \sqrt{\varrho_{i,k}}}{\sum_{i=1}^N |N_i^{max} R} \quad (50)$$

$$\sigma(\%) = \mathcal{R}_{ij} \cdot \frac{\sqrt{\sum_1^M E(\varrho_{i,k} - \mu)^2} \times 100}{\sum_1^M \varrho_{i,k}} \quad (51)$$

D. Transmission range vs localization error

The UWB nodes have one-way transmission, and the beacon nodes have two-way transmission [26], [27]. The outer and inner area of region must be same in two-level transmission, and beacon nodes must satisfy the following relation.

$$Tr(dist) = \sqrt{\frac{\dot{b}_{pl}}{\sum \forall [pl]}} \dot{\varrho}_i [\dot{b} \mapsto \dot{n}] \quad (52)$$

where $Tr(dist)$ is the total transmission distance, pl is the total number of power levels, and $\dot{b}_{pl}$ is the power sequence of each beacon node. If $Tr(dist)$ of a beacon is 50m, then $\dot{b} \mapsto \dot{n} = 50m$ therefore, the transmitted signal $\sum \forall [pl] = 2$ within a range of 36.2 m and 45 m respectively. The transmission range is not considered without path loss, that is

$$P_{RX} = P_{TX} \cdot G_{RX} \cdot G_{TX} \left(\frac{\lambda}{4\pi \varrho_i} \right)^2 \quad (53)$$

where P_{RX} and P_{TX} are the received and transmitted powers, whereas G_{RX} and G_{TX} are the gains of the receiver and transmitter. Most power loss at higher frequencies, optimal energy transfer at specified antenna gains at lower frequencies. The maximum power of the transmitted beacon is:

$$P_{R,max} = \frac{\varpi \cdot P_{T,max}}{(\varrho_i[max])^\eta} \quad (54)$$

The power computation for $\dot{b}_{pl}$ is modeled as follows:

$$\varrho_i[max] = \left(\frac{\varpi \cdot P_{T,max}}{P_{R,max}} \right)^{\frac{1}{\eta}} \quad (55)$$

$$\frac{(\varrho_i[max])^\eta}{P_{T,max}} = \frac{\varpi}{P_{R,max}} \Rightarrow \Psi(t)[ToA] \quad (56)$$

$$\varrho_{i,\dot{b}_{pl}} = \left(\frac{\varpi \cdot P_{T,\dot{b}_{pl}}}{P_{R,max}} \right)^{\frac{1}{\eta}} \Psi(t) \quad (57)$$

$$(\varrho_{i,\dot{b}_{pl}})^\eta = \left(\frac{\varpi \cdot P_{T,\dot{b}_{pl}}}{P_{R,max}} \right) \Psi(t) \quad (58)$$

from Eq. 56 and 58 we get:

$$P_{T,max} = \left(\frac{\varrho_{i,\dot{b}_{pl}}}{\varrho_{max}} \right) \cdot P_{T,max} \quad (59)$$

To compute the RMSE, a total distance of $60 \times 80m$ with 25% beacons were considered. Due to fixed LQI the accuracy increased in CL as shown in Fig. 6. The accuracy becomes stables at $\sim 1.2 \text{ m}$ over a transmission range of 70 m.

E. Accuracy improvement of noisy data using EKF

The noisy distance coordinates are refined by EKF. To overcome noise in RSS distribution, [28] presented a novel idea using quadrilaterals and clusters. In our system, we use EKF to linearize LQI values from nonlinear data observations.

TABLE II: Average LQI values from beacons to each nodes.

LQIs	B_1	B_2	B_3	B_4	B_5	B_6	B_7
N_1	117034	87.92	100.23	83.16	80.15	85.97	92.5
N_2	97.13	133.10	107.07	97.56	90.27	89.60	96.52
N_3	78.40	94.84	102.63	146.42	149.63	104.96	102.52
N_4	89.59	92.85	116.79	108.93	109.55	133.06	125.36
N_5	111.2	98.3	108.36	116.93	99.63	102.25	85.63

TABLE III: Estimated position with Centroid localization with fixed LQI as threshold

Actual Node Position				Average Condition				LOS Condition				NLOS Condition			
Nodes	x	y	z	x	y	z	Error	x	y	z	Error	x	y	z	Error
N_1	3.00	11.00	9.00	3.00	8.00	7.00	3.00	3.00	8.00	6.00	3.46	0.00	10.0	6.0	3.16
N_2	2.00	3.8	4.0	4.0	3.8	2.9	2.01	4.00	4.88	4.7	3.27	2.6	2.99	3.5	2.8
N_3	9.0	3.0	4.0	8.75	3.7	4.9	2.64	7.4	3.3	0.9	1.9	9.0	1.5	2.8	2.4
N_4	9.20	5.60	8.75	8.6	3.63	4.5	2.09	7.5	3.3	2.5	1.72	8.5	5.75	4.36	2.92
N_5	7.6	6.5	4.9	6.9	4.63	5.9	1.99	6.9	4.3	5.9	1.63	7.6	6.75	3.96	2.96

TABLE IV: Estimated position with weighted localization in LOS and NLOS condition

Actual Node Position				Average Condition				LOS Condition				NLOS Condition			
Nodes	x	y	z	x	y	z	Error	x	y	z	Error	x	y	z	Error
N_1	3.00	11.00	9.00	1.36	5.30	7.00	2.9	2.9	6.9	3.6	2.6	1.6	3.2	3.0	1.6
N_2	2.00	3.8	4.0	2.5	4.3	2.9	1.5	2.6	3.6	3.3	2.8	1.2	1.9	2.6	2.2
N_3	9.0	3.0	4.0	1.68	6.3	4.9	1.6	7.4	2.8	2.2	1.6	3.6	1.5	2.80	1.9
N_4	9.20	5.60	8.75	1.9	1.6	3.6	2.55	3.69	2.69	1.9	1.99	3.6	2.65	2.9	2.3
N_5	7.6	6.5	4.9	2.3	1.9	4.2	1.6	3.2	4.3	3.6	1.5	3.65	4.3	1.9	1.9

TABLE V: Calibration factor $\mathfrak{R}_{i,j}$ for beacon $\dot{b}$ and nodes $\dot{n}$.

G_1		G_2		G_3		G_4		G_5	
B_1	N_1	B_2	N_2	B_3	N_3	B_4	N_4	B_5	N_5
4.9	9.6	0.8	1.1	-6.9	-2.3	0.0	9.3	0.0	1.1
4.0	2.3	4.0	1.3	0.2	-2.3	0.0	2.0	0.0	0.8
2.2	3.6	2.5	2.1	2.2	2.4	3.6	0.0	2.1	0.0
0.0	2.3	3.1	2.3	0.3	1.6	1.6	0.0	9.6	0.0
0.0	1.6	1.9	1.2	-9.3	1.9	4.3	0.3	4.6	3.0
0.0	-2.6	5.6	1.3	2.5	1.3	-0.5	0.2	-4.3	1.1

TABLE VI: Estimated position with proposed localization algorithm with calibration factor in LOS and NLOS condition

Actual Node Position				Average Condition				LOS Condition				NLOS Condition			
Nodes	x	y	z	x	y	z	Error	x	y	z	Error	x	y	z	Error
N_1	3.00	11.00	9.00	0.22	5.6	6.00	1.9	2.1	6.2	0.98	1.6	1.3	2.3	2.3	0.36
N_2	2.00	3.8	4.0	2.11	6.3	2.94	1.5	2.5	5.2	1.63	0.65	1.25	0.63	1.63	0.99
N_3	9.0	3.0	4.0	7.5	4.3	1.63	0.9	6.3	1.96	2.3	1.21	2.52	1.63	1.2	1.2
N_4	9.20	5.60	8.75	1.9	4.33	2.6	0.63	1.63	2.3	1.44	0.69	2.5	2.6	1.3	0.9
N_5	7.6	6.5	4.9	2.1	1.4	1.36	1.98	2.6	3.6	3.62	1.1	2.36	1.36	0.95	0.63

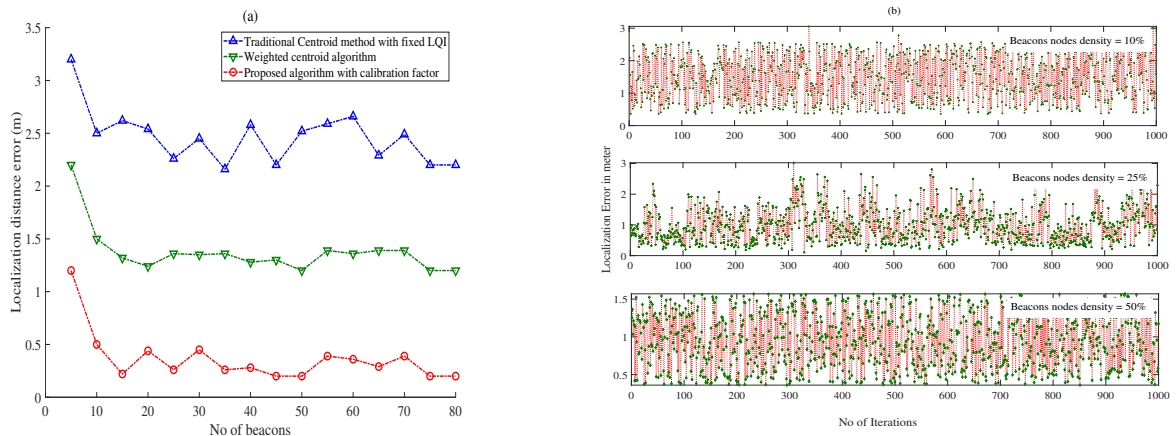


Fig. 5: (a): Comparing RMSE-SNR with diferent ToA in LOS & NLOS situation. (b): LE in varying beacon nodes density

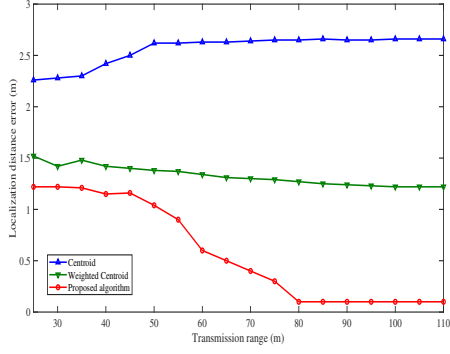


Fig. 6: Transmission range versus localization accuracy.

1) *EKF modelling*: The noise under LOS and NLOS scenarios was modeled using Eqs. (4)–(7):

$$\eta_{i,j} - \eta_{LOS} = (0, \|\varrho_i\| \cdot \mathcal{N}(\%)) \quad \text{for } LOS \quad (60)$$

$$\eta_{i,j} - \eta_{NLOS} = (\|\varrho_i\|_{min}, \|\varrho_i\|_{max} \cdot \mathcal{N}(\%)) \quad \text{for } NLOS \quad (61)$$

rand function is used to add noise in distance. EKF is an efficient framework that define localization problem by linearizing nonlinear distance estimation using first-order Taylor series consisting of a *time* and a *measurement* update.

Time update:

$$\hat{\mathbf{X}}_{k|k-1} = \mathbf{F}\hat{\mathbf{X}}_{k-1|k-1} + \mathbf{u}_k \quad (62)$$

$$\mathbf{P}_{k|k-1} = \mathbf{F}\mathbf{P}_{k-1|k-1}\mathbf{F}^T + \mathbf{Q}_{k-1} \quad (63)$$

Measurement update:

$$\mathbf{K}_k = \mathbf{P}_{k|k-1}\mathbf{H}_k^T(\mathbf{H}_k\mathbf{P}_{k|k-1}\mathbf{H}_k^T + R_k)^{-1} \quad (64)$$

$$\hat{\mathbf{X}}_{k|k} = \hat{\mathbf{X}}_{k|k-1} + \mathbf{K}_k[z_k - \mathbf{h}(\hat{\mathbf{X}}_{k|k-1}, x_k^r, y_k^r)] \quad (65)$$

$$\mathbf{P}_{k|k} = \mathbf{P}_{k|k-1} - \mathbf{K}_k\mathbf{H}_k\mathbf{P}_{k|k-1} \quad (66)$$

where $\hat{\mathbf{X}}_{k|k-1}$ and $\mathbf{P}_{k|k-1}$ are the prediction state vector and prediction error for a noisy matrix; $(\dots)^{-1}$ represents an inverse matrix operation; $\hat{\mathbf{X}}_{k|k}$ and $\mathbf{P}_{k|k}$ are the state vector and error, respectively; and $\mathbf{K}_k$ and $\mathbf{H}_k$ is the Kalman gain and Jacobian matrix of the distance measured through

$$\begin{aligned} \mathbf{H}_k &= \frac{\partial \mathbf{h}(\mathbf{x}_k), x_k^r, y_k^r}{\partial \mathbf{x}_k} \Big|_{\mathbf{x}_k = \hat{\mathbf{x}}_{k|k-1}} \\ &= \left[\frac{\hat{x}_{k|k-1} - x_k^r}{\sqrt{(\hat{x}_{k|k-1} - x_k^r)^2 + \hat{y}_{k|k-1} - y_k^r^2}}, \right. \\ &\quad \left. \frac{\hat{y}_{k|k-1} - y_k^r}{\sqrt{(\hat{x}_{k|k-1} - x_k^r)^2 + \hat{y}_{k|k-1} - y_k^r^2}} \right] \end{aligned} \quad (67)$$

The derivation of the measured distance is in the form of a nonlinear optimization. The coordinates of UWB nodes are refined using EKF in three phases as follows.

Initialization State: A KF is initialized without any assumptions regarding the initial state.

$$\begin{cases} x_k = f(x_{k+1}) + \varnothing_{k-1} \\ x_k = f(x_{k+1} \circ \alpha_k + \varnothing_{k-1} \cdot \alpha_k) \\ z_k = h(x_k) + \downarrow_k \\ z_k = h(x_k) \cdot \alpha_k + \downarrow_k \cdot \alpha_k \end{cases} \quad (68)$$

where $\varnothing_{k-1}$ denotes the noise factor of the covariance matrix from Eqs. (23) and (24).

Prediction State: The reference distance from UWB nodes to a tag is considered, with the prediction that LQI and ToA are initially recorded in a matrix, that is, $k = f(T\omega) + \mathbf{W}_{k-1}$.

Update State: After having all variables, a node vector is obtained from Eq. (68).

$$\begin{cases} \hat{\mathbb{K}}_k = z_k - h(\hat{x}_k) \cdot \zeta(t) \\ \mathcal{V}_k = \mathbf{H}_k \cdot \mathbf{P}_{k|k-1} \cdot \mathbf{H}_k^T + R_k^T \cdot \zeta(t) \\ \hat{x}_{k|k-1} = \hat{x}_{k|k-1} \cdot \mathbf{H}_k \cdot \hat{\mathbb{K}}_k \\ \mathbf{P}_k = (l_{\varrho_{i,j}} - PDF | \eta ToA \end{cases} \quad (69)$$

The noisy distribution, distance, and LQI values are now part of the update state. Here $\hat{\mathbb{K}}_k$ can be computed through the distance observations and LQI values, that is, $\mathbb{K}_k = [\varrho_{b1,\zeta(t)}, \varrho_{b2,\zeta(t)}, \dots, \varrho_{bn,\zeta(t)}]$.

$$\begin{bmatrix} \zeta(t), \hat{\mathbb{K}}_{k1} & b1 \\ \zeta(t), \hat{\mathbb{K}}_{k2} & b2 \\ \dots & \dots \\ \zeta(t), \hat{\mathbb{K}}_{kn} & bn \end{bmatrix} [\Psi(t)] \quad (70)$$

$$\therefore \mathbf{H}_k = \begin{bmatrix} \varrho_{i,j} \left[\frac{\partial \mathbf{h}(\mathbf{x}_k), \zeta(t), \hat{\mathbb{K}}_{k1}}{\mathbb{K}_{k1} \mathbf{x}_{k1}} \right] & \varrho_{i,j} \left[\frac{\partial \mathbf{h}(\mathbf{y}_k), \zeta(t), \hat{\mathbb{K}}_{k2}}{\mathbb{K}_{k2} \mathbf{y}_{k2}} \right] \\ \dots & \dots \\ \varrho_{i,j} \left[\frac{\partial \mathbf{h}(\mathbf{x}_k), \zeta(t), \hat{\mathbb{K}}_{kl}}{\mathbb{K}_{kl} \mathbf{x}_{kl}} \right] & \varrho_{i,j} \left[\frac{\partial \mathbf{h}(\mathbf{y}_k), \zeta(t), \hat{\mathbb{K}}_{kl}}{\mathbb{K}_{kl} \mathbf{y}_{kl}} \right] \end{bmatrix} \quad (71)$$

$$\mathbf{T} = \begin{bmatrix} \mathbf{H}_k \cdot \sigma_x^2 & 0 & 0 & 0 & 0 \\ 0 & \mathbf{H}_k \cdot \sigma_y^2 & 0 & 0 & 0 \\ 0 & 0 & \mathbf{H}_k \cdot \sigma_z^2 & 0 & 0 \\ 0 & 0 & 0 & \mathbf{H}_k \cdot \sigma_{\varnothing_{k-1}}^2 & 0 \\ 0 & 0 & 0 & 0 & \mathbf{H}_k \cdot \sigma_{\mathbb{K}_k}^2 \end{bmatrix} \quad (72)$$

Noise is reduced, and the matrix in Eq. (24) is updated. A flow diagram of the EKF framework is shown in Fig. 7.

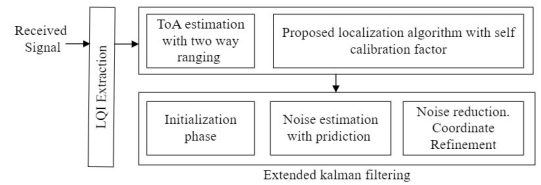


Fig. 7: EKF structure for proposed localization scheme.

To analyze the accuracy of EKF and LE, 10 beacons and 15 nodes are deployed on a $150\text{m} \times 70\text{m}$ area, with 5000 trials, covering a total area of $5000 \times (150 \times 70)\text{m}$ with 5000 beacons and 5000×15 nodes. During simulation, an overall error of

0.2m is recorded. The LE decreased with EKF framework after multiple iterations, as shown in Fig. 8. By combining EKF with naïve noise, the CF weight becomes constant, leads to high LE. With 15 beacons and 10 UWB nodes, the Gaussian noise provides an error of 0.23m. The CF is compared with diverse noises using EKF in the table VII and Fig. 9.

F. Lower Bound (LB) of the Location Error

Consider the following relationship:

$$lb_{\dot{b}, \dot{n}} = \dot{b}_0 - \eta \cdot \ln \frac{\varrho_{i,j}}{\varrho_0} + \zeta(t), \quad (73)$$

where $\varrho_0 \neq 0$, we set $\varrho_0 = \Re_{ij}$. From the above relation, the n number of LQI measurements obtained from the four reference points $[\dot{b}_0, \dots, \dot{b}_n]$ can be expressed as:

$$lb_n = \dot{b}_0 \cdot 1 - \eta \dot{b}_{x,y,z} + \zeta(t) \quad (74)$$

For the node position, the LB distribution is obtained by $\dot{n}_n = [\dot{n}_{1,n}, \dots, \dot{n}_{4,n}]^T$ with a Gaussian noise vector distribution $\zeta_{x,y,z} = [\varrho_1, \dots, \varrho_4]^T$. All four unknown parameters were transformed into a vector $\theta = [(x, y, z)_0, \sigma^2]^T$ and the Fisher information matrix (FIM) for each obtained LQI was modeled.

$$G(FIM) = \frac{\eta^2}{\sigma^2} \begin{bmatrix} \frac{\partial \dot{b}}{\partial \dot{b}_0} & \frac{\partial \dot{b}}{\partial \dot{b}_0, \partial \dot{b}_1} & \frac{\partial \dot{b}}{\partial \dot{b}_1, \partial \dot{b}_2} & 0 & 0 \\ \frac{\partial \dot{b}}{\partial \dot{b}_0, \partial \dot{b}_1} & \frac{\partial \dot{b}}{\partial \dot{b}_1, \partial \dot{b}_2} & \frac{\partial \dot{b}}{\partial \dot{b}_2, \partial \dot{b}_0} & 0 & 0 \end{bmatrix} \quad (75)$$

which can deviate easily from Eq. (72). Thus, the cramer-rao bounds of the EKF coordinates of the estimated positions are

$$CRB_{x_0} = \frac{1}{N} \| G(FIM)^{-1} \| [0, y, z] \quad (76)$$

$$CRB_{y_0} = \frac{1}{N} \| G(FIM)^{-1} \| [x, 0, z] \quad (77)$$

$$CRB_{z_0} = \frac{1}{N} \| G(FIM)^{-1} \| [x, y, 0] \quad (78)$$

$$\therefore \Rightarrow Eq.41 : \varrho_l \| \dot{b} - \dot{b} \|^2 = \varrho_{i,k} \geq [CRB_{x_0} + CRB_{y_0} + CRB_{z_0}] \quad (79)$$

where $[CRB_{x_0} + CRB_{y_0} + CRB_{z_0}]$ is LB of the LE.

G. Comparison with state-of-the-art techniques

A performance comparison of the proposed technique with state-of-the-art results. A self-calibrated localization solution for sports applications implemented using UWB technology was presented by [29]. This technique relies on Decawave DW1000 devices that measure the ToF, outlier identification, and anchor position through multilateration, which also supports a two-way ranging system with a communication radius of 60 m. For experimental purposes, eight anchor nodes were deployed at 1.9 m along with their known relative distances. The CDF of the tag localization error even exceeded 3 m after calibration. In the proposed technique, it was only 0.6 cm initially, and reduced to 0.23cm with sufficient beacon. The CF also helps to overcome localization errors, even in

the presence of a “Dumb” node in a certain region. Another method of self-calibrated UWB localization was proposed in [30], which computes the number of anchor nodes, accuracy in NLOS environment, and design system precision. Therefore, the error at a certain point is difficult to manage and is higher than 6.9 m. The overall localization accuracy and simulation results are presented in Table VIII and Figures 10.

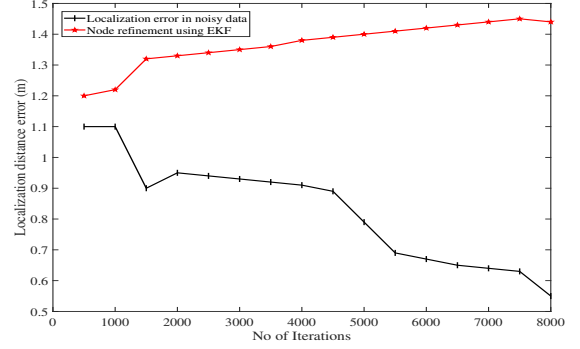


Fig. 8: Localization error accuracy using EKF framework.

TABLE VII: Localization error (m) + AWGN and EKF.

Nature of Experiment	$\dot{b} = 15$	$\dot{b} = 25$
Classic proposed Algorithm	0.59	0.33
Proposed Algorithm + Naive weighting noise	2.62	2.15
Proposed Algorithm + Intelligent weighting noise	2.66	2.5
Proposed Algorithm + Gaussian noise	1.98	1.38
Proposed + EKF + Naive noise	1.32	0.98
Proposed + EKF + Intelligent noise	1.18	0.66
Proposed Algorithm + EKF + Gaussian Noise	1.12	0.23

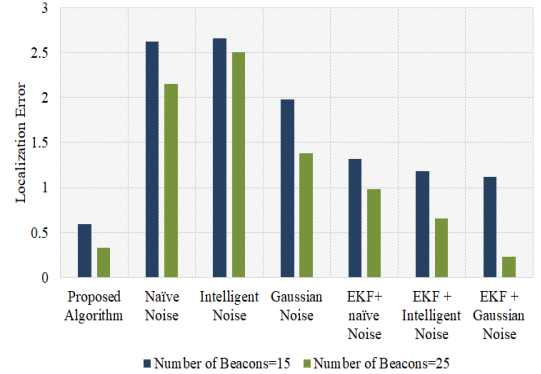


Fig. 9: MSE of proposed algorithm in different types of noises.

Ref	Year	Title	Localization Error (m)
[32]	2022	Self-Calibrated localization for sport application	~1.9-3
[33]	2022	Indoor positioning with modified LLS	~6.9
Proposed	-	Self-calibration localization under NLOS	~0.23-0.4

TABLE VIII: Comparison of the proposed technique with state-of-the-art methods

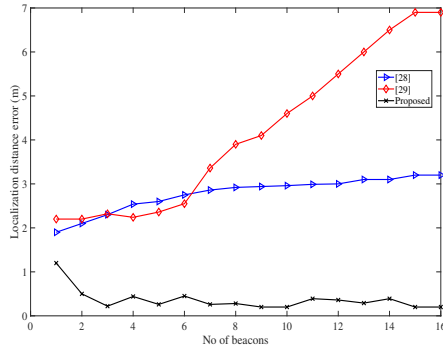


Fig. 10: Comparison of LE with existing techniques.

VII. CONCLUSION

A novel UWB localization using self-calibration is proposed that utilize LQI instead of RSSI for context-aware applications. Our approach employs ML estimators for ToA based on first signal path detection, prioritizing accuracy and network coverage in challenging conditions. Proposed solution addresses the calibration challenges in weighted-based localization techniques during link failure. The algorithm measures LQI values at gateways, relieving nodes from processor and battery-intensive tasks. The presence of noise and MPCs affect ToA estimation and LE. Our algorithm, tested with various noise types, showed improved accuracy with EKF, refining node coordinates. Numerically, our research demonstrates remarkable results. In full network coverage and noisy environments, the proposed solution achieves a 0.23m error reduction, enhancing overall accuracy and resilience to environmental changes

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